Cosmology with peculiar velocities of galaxies in the nearby Universe

Proposal overview

The goal of COSMOPV is to investigate the accelerated nature of the expansion of our Universe by producing robust cosmological measurements with state-of-the-art observations. We aim at measuring the growth rate of structures, quantity that can probe dark energy models and deviations from general relativity on cosmic scales.

The team will use galaxy and supernova datasets from the Dark Energy Spectroscopic Instrument (DESI) and the Zwicky Transient Facility (ZTF), producing the best density and peculiar velocity fields in the Northern sky.

COSMOPV is divided in three major components:

1. **Simulations:** producing large samples of sythetic versions of DESI and ZTF surveys;
2. **Systematics:** quantifying the impact of observational systematics in real and synthetic datasets;
3. **Cosmology:** obtaining cosmological constraints on dark energy and modified gravity theories.

The optimal and unique combination of DESI densities and DESI+ZTF velocities is expected to yield 8.3% precision on the growth rate at late cosmic times. Combined with other growth rate measurements, COSMOPV's results have the potential to reduce uncertainties on the gravitional index γ by a factor of two.

Our local team has the required expertise and has acquired international recognition in this field over recent years, and will work hand-in-hand with experts from DESI and ZTF collaborations.

The success of COSMOPV will critically depend on the workforce in our team required to merge efforts across datasets and collaborations. Therefore we request funds to hire one PhD student and one experienced postdoctoral researcher to achieve the objectives of COSMOPV.

COSMOPV's team members

Last Name	First name	Current position	Role and responsibilities in the project	Involvement (person.month)
Bautista	Julian	Professor	Coordinator All tasks	43.2
Fouchez	Dominique	Researcher	Work package N Provide x and y	4.8
Racine	Benjamin	Researcher	Work package N Provide x and y	4.8
Feinstein	Fabrice	Professor	Work package N Provide x and y	19.2
Mena-Fernandez	Juan	Postdoc	Work package N Provide x and y	14.4
Hanser	Corentin	Postdoc	Work package N Provide x and y	14.4
Kebadian	Rafael	PhD student	Work package N Provide x and y	14.4
To be recruited		PhD student	Work package 1 Blabla Work package 2 Blabla	36
To be recruited		Postdoctoral researcher	Work package 1 Blabla Work package 2 Blabla	36

I. Proposal's context, positioning and objective(s)

a. Objectives and research hypotheses

The accelerated nature of the expansion of our Universe remains one of the most mysterious puzzles in Physics, well consolidated by several recent and independent cosmological observations [1–5]. Explanations for such an acceleration require Physics beyond our standard model and/or ambitious modifications of our current theory of gravity: general relativity.

One of the potential solutions is to introduce a new exotic component of our Universe: **dark energy**. Dark energy has negative relativistic pressure, therefore acting as a source of anti-gravity, repelling galaxies on large cosmic scales. The simplest form of dark energy is equivalent to Einstein's cosmological constant Λ. The cosmological constant corresponds to a fluid with an equation of state $w = -1$ (relation between relativistic pressure and energy density), constant with time. Current observations require that at least 68% of our Universe's energy budget is in form of dark energy. Such a model is dubbed ΛCDM, where CDM stands for cold dark matter. More exotic dark energy models predict different values for w, with the only constrain that $w < -1/3$ to produce an accelerated expansion. A generalised model considers the equation of state of dark energy to be an time-evolving quantity [6, 7], which can be written as $w(a) = w_0 + w_a(1-a)$, where a is the scale factor of the Universe ($a = 1$ today and $a = 0$ at the Big Bang) and w_0, w_a two fixed parameters. Such a model is referred to as w_0w_aCDM. In 2024, a combination of several cosmological measurements (see below) showed strong preference for an w_0w_aCDM Universe over a ΛCDM one by more than 3σ [5], showing hints of new Physics for the first time since the discovery of the acceleration in the 2000's.

An alternative solution to explain the accelerated expansion of the Universe is to propose alternatives to general relativity (GR) to describe gravity on large cosmic scales. Those are referred to as **modified gravity theories**. Modified gravity effects have to manifest only on cosmic scales where the expansion is observed, since GR is compatible with extremely high-precision measurements made on Earth [8–12] in the Solar System [13, 14], on stars [15, 16], on binary and triple systems of pulsars [17–20], as well as in mergers of black holes and neutron stars [21, 22]. Direct imaging of black holes also are consistent with GR, in a very strong field regime [23]. There are several families of models trying to extend or modify GR to account for the observed acceleration of the Universe [24–28]. However, such models predict the same *expansion history* as dark energy, so we need an alternative observable to discriminate between dark energy and modified gravity models. One of the best observables is the *growth history* of the cosmic density field of the Universe and its impact on its velocity field.

The expansion rate of the Universe $H(z)$ and **the growth rate of the structures** $f(z)$, both functions of cosmic time or redshift z, are two important observables that jointly allow us to constrain and distinguish between dark energy or modified gravity models [28, 29].

The expansion rate of the Universe, $H(z)$, as a function of time or redshift z, has been historically been measured via the average distance-redshift relationship of extragalactic objects. Redshifts are precisely measured with spectroscopy, but distances are a more challenging endeavour. Several distance indicators have been used in recent years: the Tully-Fisher (TF, [30]) and Fundamental Plane (FP, [31]) relations, as well as type-Ia supernovae (SNIa, [32]). The first two indicators (TF and FP) rely on empirical relations between galaxy properties and their luminosity or size, so in principle they can be applied larg samples of galaxies. However those relations have a large intrinsic scatter of about 20-30% in distances compared to SNIa (5-10%). SNIa are better distance indicators but are rarer events. The discovery of the accelerated expansion in the 2000's used samples of tens of SNIa [33, 34] while today's samples reach several thousands (see below). **COSMOPV will employ state-of-the-art samples of distance indicators to constrain models of dark energy and modified gravity.**

The growth rate of structures $f(z)$ quantifies how fast initial perturbations of the cosmic density field, $\delta(\vec{x})$, grow with time or redshift z. The r.m.s. of the cosmic density field smoothed on scales of $8h^{-1}$ Mpc noted $\sigma_8(z)$ also quantifies structure growth versus time. The product $f(z)\sigma_8(z)$ can be measured via the impact of *peculiar velocities* of galaxies on the observed structures. Peculiar velocities are simply due to the motions of galaxues due to gravitational attraction from dense structures, therefore those velocities are an important laboratory to test gravity on cosmic scales. Peculiar velocities (PV) modify redshifts due to the cosmic expansion. Both are

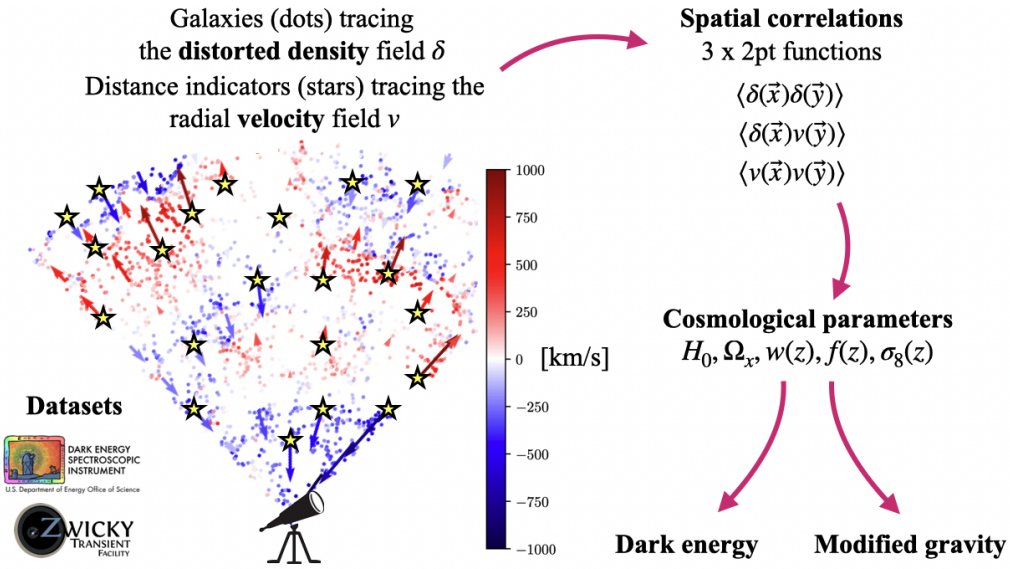

Figure 1. Schematic view of the COSMOPV. We will use state-of-the-art samples of galaxy redshifts and distance indicators to measure both the distorted density field and the radial peculiar velocity field to derive better constraints on the growth rate of structures, dark energy and modified gravity theories.

generally undistiguishable, creating distortions in the observed galaxy density field, known as *redshift-space distortions* (RSD). The two-point statistics of the galaxy density field, $\langle\delta(\vec{x})\delta(\vec{y})\rangle$, becomes anisotropic w.r.t. the line-of-sight, allowing measurements of the growth rate $f(z)$ with spectroscopic galaxy surveys [35–37]. Moreover, if distance indicators are available for a large sample of galaxies, it is possible to break the degeneracy between velocity due to cosmic expansion and velocity due to proper motion. Measuring the velocity field $v(\vec{x})$ brings extra information than just the distorted galaxy density field $\delta(\vec{x})$. The two-point statistics of the velocity field, $\langle v(\vec{x})v(\vec{y})\rangle$, also depends on $f(z)\sigma_8(z)$. Measuring three combinations of two-point functions, i.e., density-density, velocity-velocity and their cross correlation $\langle\delta(\vec{x})v(\vec{y})\rangle$, yield tighter constraints than just one of such statistics [38–41]. Figure 1 shows a schematic view of such a combination. **COSMOPV will employ state-of-the-art samples of distance indicators (FP, TF, and SNIa) to measure peculiar velocities and obtain the tightest constraints on the growth rate at late times.**

b. Position of the project in relation to the state of the art

The COSMOPV team has more than five years of expertise in this field and it is currently well established as leads in key cosmological analyses with peculiar velocities, including surveys such as:

- **the Dark Energy Spectroscopic Instrument (DESI):** currently the best spectroscopic survey of galaxies to date, it will measure 63 million galaxy redshifts after eight years of observations (2021-2028) over 17,000 deg^2 [42]. DESI galaxies are an excellent tracer of the distorted density field, from which we can measure anisotropies due to RSD. The DESI Peculiar Velocity survey will produce distance measurements to 53k spiral galaxies from the Tully-Fisher relation and 133k ellipticals from the Fundamental Plane [43],Those distances will yield a well sampled velocity field at $z < 0.1$.
- **the Zwicky Transient Facility (ZTF):** a photometric survey with large field of view and high cadence, ideal for the discovery of transient events, including SNIa [44]Covering more than 17k deg^2 of the Northen sky, perfectly overlapping with DESI observations, ZTF will discover more than 30k SNIa over redshifts $0 < z < 0.2$. which will be used to obtain precise peculiar velocities.

COSMOPV will focus in the DESI and ZTF overlap, a perfect opportunity for cross-correlations and systematic cross-checks. Based on our own Fisher forecasts [45, 46], **COSMOPV will produce a 8.3% precision growth rate measurement when combining DESI galaxies and DESI+ZTF velocities (Figure 2), representing the most precise and robust estimate at late cosmic times.**

Producing large samples of synthetic datasets reproducing real data at high fidelity is an essential ingredient.

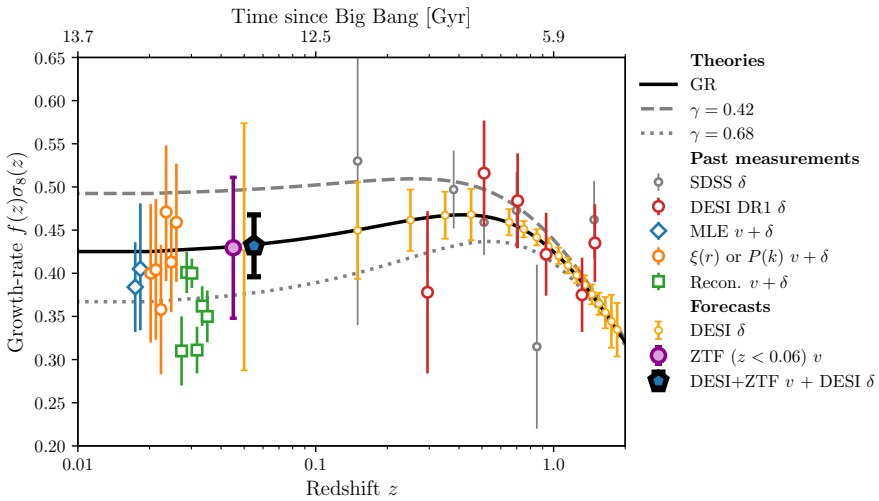

Figure 2. Growth rate of structures versus redshift from past measurements compared to expected results from COSMOPV, which will employ low-redshift galaxies and peculiar velocities from the Dark Energy Spectroscopic Instrument (DESI) and the Zwicky Transient Facility (ZTF).

Those sets have multiple uses: testing our methodologies for biases, understand our uncertainties, compute covariance matrices of observables, and study the impact of systematic effects on cosmological results. Building up on current experience in our team [45, 47–51], **COSMOPV will lead the production of the most realistic synthetic datasets for both DESI and ZTF.**

c. Methodology and risk management

Describe precisely the methodology and its relevance to reach the objectives; detail the scientific risks and fallback solutions envisaged; describe the scientific program and justify the breakdown of the work program into tasks in line with the objectives pursued.

- For each task, describe the objectives, the work programme, deliverables, contributions of the partners, methods and technical choices, risks and fallback solutions (among other examples: difficulty to access study site, lack of preliminary data, delay in obtaining approval from the ethics committee, etc.). Illustrate with a Gantt chart.
- Justify the relevance of the methodology in terms of ethics, scientific integrity and social responsibility of sciences - and as such, considering or not the sex and/or gender aspect , considering the ecological sustainability in the implementation of the project -, and in terms of disciplinary coverage (mono- trans- and inter-disciplinarity).
- If applicable, indicate the conditions of access to a research infrastructure IR, IR* or OSI

COSMOPV is organised in three connected work packages: synthetic datasets, systematic effects in real data, and cosmological implications of our results.

Work package 1: Synthetic datasets

WP1 will focus on the production and analyses of realistic synthetic datasets reproducing properties of real data from DESI and ZTF.

- **Fast approximate simulations** Recent mock catalogs used for peculiar velocities are based on n-body simulations such as AbacusSummit [51]or Uchuu [52], which are high accuracy but are limited in volume or in the number of independent realisations. Using well-known techniques [48, 53], we will produce a large number ($O(1000)$) of independent realisations of density and velocity fields at low redshift.
- **Production of synthetic DESI Peculiar velocity surveys and ZTF SNIa samples:** We will add survey-like properties to the simulations above to create sythetic samples of DESI and ZTF surveys, including distance indicators. For DESI, we will use existing infrastructure used to produce DESI Data Release 1 mock catalogues for TF and FP samples [51] while for ZTF, we will build upon in-house experience [45] and use existing software (SKYSURVEY) from the collaboration. Those new mock catalogues will reach higher redshifts than current sets, and will include subtle effects of redshift evolution and selection effects.

- **Cross-covariance between methodologies and cosmological probes:** Several methodologies are able to exploit a distorted density field and a peculiar velocity field (e.g. [41, 46]). Those are commonly applied to the same initial dataset, so results are correlated. Using large sets of mock catalogues, we will precisely estimate this cross-correlation, which is key to derive consensus results (e.g. [54–56]). The same mocks could be further improved to evaluate the cross-covariance between growth measurements from peculiar velocities and weak lensing.

Work package 2: Systematic effects in real data

Focusing on real data, WP2 will investigate potential systematic effects affecting growth measurements and will yield the final combined result from DESI and ZTF.

- **Quantify the impact of observational systematics:** With the mocks from WP1, we will quantify for the first time the impact of several observational systematics to an unprecedented level of accuracy. Several systematic effects can create spurious correlations in the observed velocity field, potentially biasing our growth rate measurements [57]. We will pay particular attention to the angular dependence of the Malmquist bias, dust attenuation, flux calibration residuals, among others. Strategies will be put in place to mitigate such impacts when possible, otherwise a systematic uncertainty budget will be propagated to final results.
- **Growth rate measurement with a photometric sample of SNIa:** Building up on our experience with synthetic datasets of Rubin LSST SNIa [50, 58], we will extend this work to the final release of ZTF SNIa data. This photometrically-classified sample can potentially increase the number of SNIa by factors of 5 to 6 compared to the spectroscopically-classified sample, also increasing the redshift range to $0 < z < 0.2$.
- **Cross-calibration of DESI and ZTF peculiar velocities:** Given the existence of three different distance indicators (TF, FP and SNIa) in the same volume, there are many opportunities for cross-calibrations and consistency checks between those, in regards to cosmological parameters. Cross-correlations between different surveys are much less sensitive to systematic effects, so measuring them adds to the robustness of our analysis. We aim at obtaining a final consensus growth rate measurement from this combination, including all contributions of potential systematic errors.

Work package 3: Cosmological implications of our measurements

WP3 will derive cosmological constraints from results of WP1 and WP2. Our growth rate measurement at low redshift, when combined with high redshift expansion and growth rate measurements (Fig. 2), has the potential to improve constraints on dark energy and rule out some modified gravity theories.

The expansion rate will be provided by SNIa Hubble diagrams, baryon acoustic oscillations and the cosmic microwave background. At high redshift, DESI and Euclid will also provide precise growth rates from RSD. With those probes, WP3 will deliver constraints on:

- Ω_m, σ_8, H_0 when assuming a ΛCDM model
- w_0 and w_a parameters describing the equation of state of evolving dark energy
- γ or (μ_0, Σ_0) which empirically parametrise deviations from general relativity. In-house forecasts predict that the uncertainty on γ reduces by a factor of ~ 2 when including our low-redshift growth rate measurement.

Those results will set the tightest constraints from the Northen sky. This work serve as basis for future analyses of datasets in the Southern sky such as Rubin LSST and 4MOST.

II. Organisation and implementation of the project

COSMOPV is a PRME project that will be hosted at Aix-Marseille Université (amU), particularly at the Centre de Physique des Particules de Marseille (CPPM). The team consists of researchers already active in the field, all members of either DESI and ZTF collaborations, spaning a wide range of expertise.

a. The scientific team at the CPPM

1. **Julian Bautista - Scientific Coordinator (Professor, 100% ETPR)**
   Expert in galaxy clustering, type-Ia supernovae and peculiar velocities for cosmology. Major contributor of past cosmological results with spectroscopic surveys such as SDSS-III BOSS, SDSS-IV eBOSS. Currenly active member of DESI and ZTF and actively working in peculiar velocities for the past 5 years, with the support of the Chaire d'Excellence A*Midex. Current co-chair of the DESI Transients and Low-redshift Cosmology Working Group, whose main goal is to perform growth rate measurements with the DESI Peculiar Velocity Survey.
2. **Dominique Fouchez (CNRS Research Director, 10% ETPR)**
   Expert in supernova cosmology (SNLS, ZTF and LSST), photometric detection of sources as well as applications of machine learning to astronomy.
3. **Benjamin Racine (CNRS Researcher, 10% ETPR)**
   Expert in cosmology with the CMB and SNIa. Leading the efforts to homogenise the flux calibration of ZTF images to produce datasets suited for precision cosmology with SNIa. His raw data expertise is of extreme value for COSMOPV, as he will be able to advice on potential sources of systematic effects that impact cosmological results.
4. **Fabrice Feinstein (Professor, 40% ETPR)**
   He has been closely working with B. Racine on ZTF datasets and has experience in supernova cosmology. Will contribute with the investigation of observational systematics in the data.
5. **Juan Mena-Fernandez (Postdoc, 30% ETPR)**
   Expert in galaxy clustering, weak lensing and mock catalog production. Will be joining the team in Nov 2025 for three years and part of his work will be related to COSMOPV.
6. **Corentin Hanser (Postdoc, 30% ETPR)**
   Transitioning from cosmology with galaxy clusters to peculiar velocities. Will contribute during one year to the project.
7. **Rafael Kebadian (PhD student, 30% ETPR)**
   Currently leading the measurement of the growth rate of structures with half of ZTF data, will be part of the project during his last year.

Implication of the scientific coordinator in on-going projects

Name of Researcher	Person. month	Call, funding agency, grant allocated	Title of the project	Name of the scientific coordinator	Start - End
Julian Bautista	N/A	N/A	N/A	N/A	N/A

b. Implemented and requested resources to reach the objectives

Including fees, **the total budget of COSMOPV amounts to 473 697,93 € for a duration of 48 months.** It is divided between staff expenses, travel support, computers, conference fees and conference organisation, plus admin fees.

Staff expenses: 391,3k €

We would like to hire:

- one PhD student for 36 months for a cost of 137 977,00 €;
- one senior postdoctoral researcher for 36 months for a cost of 253 378,00 €.

The senior postdoctoral researcher would have more than 2 years of experience after their PhD degree.

Equipments: 6k €

Two personnal computers for the future PhD student and postdoctoral researcher, each one at the maximum cost of 3k €. Recently the price of RAM memory increased considerably, driving up the cost of new computers.

Those computers will be essential to the project as they will serve the two new members of the team to develop analysis code and acces the supercomputers (Mardec, NERSC) where the high-performance computing will be performed.

Service delivery: 3 k€

We will publish our findings in the Astronomy & Astrophysics journal (A&A). Recent changes to publication policy limit the number of pages to a maximum of 12 without publication fees. Currently, each extra page costs 150 €. Since our publications often exceed this limit, we will request 3k € for publication fees.

As a reminder, open access versions of our published articles will be posted on the arXiv website without any associated fees.

Other costs: 30,5 k€

We would like to ask **19k € for travel support** for each member of the team, including the scientific coordinator. Each new member of the team will be assisted with 2,5k € per year, while the scientific coordinator will be assisted with 1k € per year. We ask an additional 4k € for the payment of conference fees for the same people.

To increase the visibility of COSMOPV, as well as our local institutions, we would like request **7,5k € to organise a version of the CosmicFlows conference in Marseille**. Such funds will help with the costs of the conference, such as room rentals, coffee breaks and conference dinner. We expect to welcome around 50 people for a duration of a week, to share about the domain of cosmology and astrophysics with peculiar velocities.

Requested means by item of expenditure and by partner

Type of expenditure	Amount (€)
Staff expenses	391 355,00
Equipments	6 000,00
Service delivery	3 000,00
Other costs	30 500,00
Structure costs**	58 165,43
Sub-total	489 020,43
Requested funding	489 020,43

III. Impact and benefits of the project

a. Publication

Publication of results in the grade A peer-review journal Astronomy and Astrophysics (A&A). We will ensure that open-access versions of our publications will be published on the arXiv website, as required by the CNRS publication policy.

b. Presentations in cosmology conferences

Our results will be advertised in the main cosmological conferences around the world. Presentations are expected by the two new members of the group as well as the scientific coordinator. These presentations will also contribute to the exposure of COSMOPV members as they search their next scientific opportunities.

c. Organisation of a local CosmicFlows conference

During the project, we plan to organise a version of the CosmicFlows conference locally in Marseille. This conference unites experts from the field of cosmology and astrophysics with peculiar velocities. The scientific coordinator was one of the invited speakers of the last version of the CosmicFlows[1] in 2025 in Brisbane, Australia, tying up good links with their organizers.

d. Press-releases

Major results within the DESI or the ZTF collaborations can be part of a coordinated world-wide press-releases. Those press-releases are usually republished by main news papers such as Le Monde, Le Figaro, among others, significantly increasing the impact of COSMOPV to the general public.

[1] https://indico.global/event/12931/

e. Outreach events

Reaching out to the general public is one of the priorities of COSMOPVṪhe members of the team will be invited to contribute to various outreach events in town and connect to the next generation of potential researchers.

We will continue our participations to the *Fête de la Science*.

Pint of Science

Workshops for high-school students

Internships for first and second-year undergraduate students.

IV. References related to the project

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